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Quantitative estimation of minimum offset for multichannel surface-wave survey with actively exciting source

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Abstract

Multichannel analysis of surface waves is a developing method widely used in shallow subsurface investigations. The field procedures and related parameters are very important for successful applications. Among these parameters, the source–receiver offset range is seldom discussed in theory and normally determined by empirical or semi-quantitative methods in current practice. This paper discusses the problem from a theoretical perspective. A formula for quantitatively evaluating a layered homogenous elastic model was developed. The analytical results based on simple models and experimental data demonstrate that the formula is correct for surface wave surveys for near-surface applications.
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Keywords: Rayleigh waves; Multichannel analysis; Lower limit of offsets; Layered homogenous elastic model

1. Introduction

The spectral analysis of surface wave (SASW) method has been used for site investigation for several decades (e.g., Nazarian et al., 1983; Al-Hunaidi, 1992; Stokoe et al., 1994; Tokimatsu, 1995; Ganji et al., 1997). SASW method uses the spectral analysis of a surface wave generated by an impulsive source and recorded by a pair of receivers. To improve inherent difficulties in evaluating and distinguishing signal from noise with only a pair of receivers by this method, a new technique incorporating multichannel analysis of surface waves using active sources, named as MASW, was developed (Park et al., 1999b; Xia et al., 1999a). The MASW method rapidly aroused many researchers' attention (Miller et al., 1999; Xia et al., 1999b, 2002a,b,c, 2004a,b, 2005, in press; Yilmaz and Eser, 2002; Tian et al., 2003a,b; Beaty et al., 2002; Beaty and Schmitt, 2003; Liu et al., 2004; Lin and Chang, 2004; Lin et al., 2004). This method can use advanced technologies of digital signal processing for multichannel records (Dal Moro et al., 2003), and therefore take advantage of the many benefits from time–space information of the seismic wavefield. For example, it can be more flexible in discriminating and integrating fundamental and higher modes of Rayleigh waves (Xia et al., 2003). Inversion of Rayleigh waves with higher modes can stabilize inversion processing and obtain a shallow subsurface shear wave velocity profile with higher resolution (Xia et al., 2003).

Despite the merits of MASW, many practical problems result from its applications. After many site tests, we found that the source to the first receiver distance, traditionally called the nearest offset in seismic explo-

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Table 1
Suggested offset corresponding to shallow shear-wave velocities

Material type (V_S in m/s)	Very soft (V_S<100)	Soft (100<V_S<300)	Hard (200<V_S<500)	Very hard (500<V_S)
Offset (m)	1–5	5–10	10–20	20–40

(Modified from the short course notes of SurfSeis© by Exploration Services Section at KGS.).

ration, should be set shorter when a small hammer hits lightly on a striker plate in contact with the surface and longer when a large hammer hits heavily because spectra of surface waves generated with different strength of hits were different. This phenomenon is partly near-field effects and has been discussed by some researchers (e.g. Park et al., 1999a). To our knowledge, Ewing et al. (1957) was the first one who studied this problem. Based on their observation of earthquake data, they present a formula, which can be used to evaluate the nearest offset, although they did not give any further explanation. They claimed that the receiver should be set at a distance d as

$$d<\frac{V_R h}{\sqrt{V_P^2 - V_R^2}}, \; or \tag{1a}$$

$$d<\frac{V_R h}{\sqrt{V_S^2 - V_R^2}}, \tag{1b}$$

where Rayleigh waves can be recorded or observed, where V_p, V_s, and V_R represent velocities of compression (P), shear (S), and Rayleigh waves, respectively, and h is the depth of the source.

With surface impact sources used in near-surface applications, one can observe Rayleigh waves at the source point. Recorded Rayleigh waves at this setting have little use in near-surface applications because of wavefield complexity due to plasticity and failure of plane-wave approximation in the vicinity of the source. On the other hand, methods of empirically determining the optimum offset exist. One of these methods is based on wavelength of Rayleigh waves, e.g. Stokoe et al. (1994) suggested the optimum offset may be estimated by one half of wavelength. Meanwhile others developed a different rule of thumb in determining the nearest offset (Gucunski and Woods, 1991; Park et al., 1999a). Zhang et al. (2004) developed an optimized offset estimation based on the energy maximization criterion of dispersive modes. However, according to the criterion, the resolution of the dispersion image would in some cases decrease although the conditions of energy maximization of the dispersive modes were favored. The Exploration Services Section at the Kansas Geological Survey (KGS) has suggested the optimum offset should be selected according to site properties of near-surface materials and has provided an empirical chart (Table 1, modified from Short course note on SurfSeis© software).

Despite the effectiveness of the previous mentioned methods, whether or not suitable for adaptation, one would expect there should be a quantitative method, supported by theory to make source-to-closest-receiver estimations even though it may only be valid for a simplified earth model. This paper aims to access the theory and discusses requirements resulting from near-surface applications.

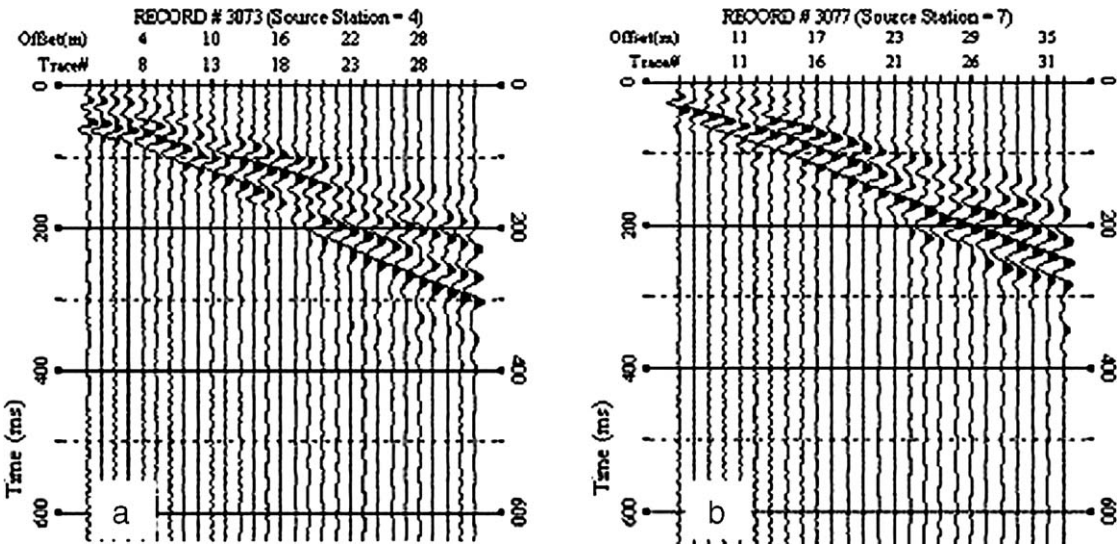

Fig. 1. The first 600 ms of shot record #3073 (a) and #3077 (b).

2. Facts from dispersion analysis of exploration data

These surface-wave data were acquired for a shallow geology mapping project in Joplin, Missouri, in 1997. The source was an IVI high-frequency vibrator (Minivib) with a 2.4-m shot spacing and a 10-s linear up-sweeping from 10 to 100 Hz. The receiver interval was 1.2 m. Out of the 544 shot gathers acquired for this project, several possessed the nearest offset of one half of the receiver interval (Fig. 1). The source was positioned at traces four and seven for shots # 3073 and # 3077, respectively. The depth from the earth surface to the bedrock is 12.0 m with a generally homogeneous overburden of unconsolidated sediments, determined from drilling records near shot # 3073 (Xia et al., 1999b). From a practical perspective, these records possessed high-quality surface-wave energy. Results of dispersion analysis on these records using SurfSeis© are shown in Fig. 2. Only the fundamental mode can be identified from the dispersive image (Fig. 2a), in which there was stronger body-wave energy between 32 to 45 Hz. However, higher

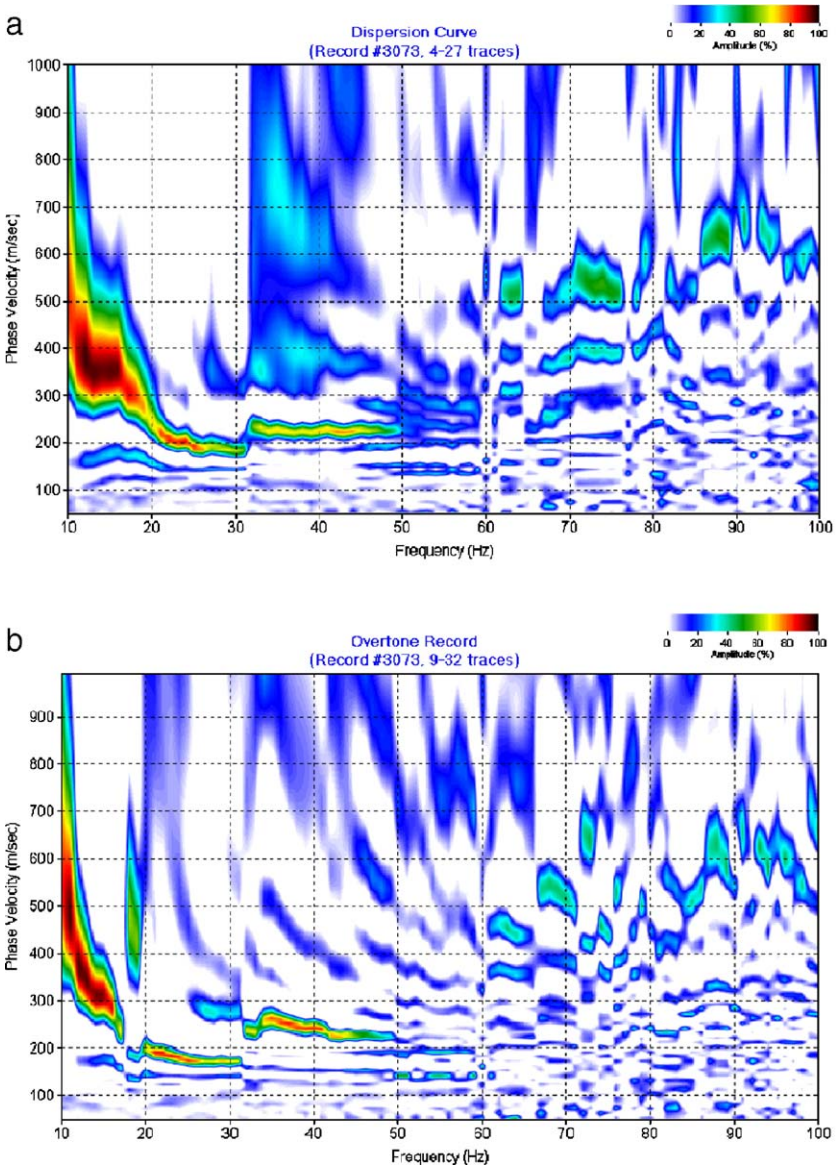

Fig. 2. Dispersion analysis of record #3073 acquired in Joplin, Missouri, in 1997: (a) based on traces from 4 to 27; (b) based on traces from 8 to 31; (c) based on traces from 4 to 8; and (d) based on traces from 9 to 13. Parameters were kept the same during generation of the previous images.

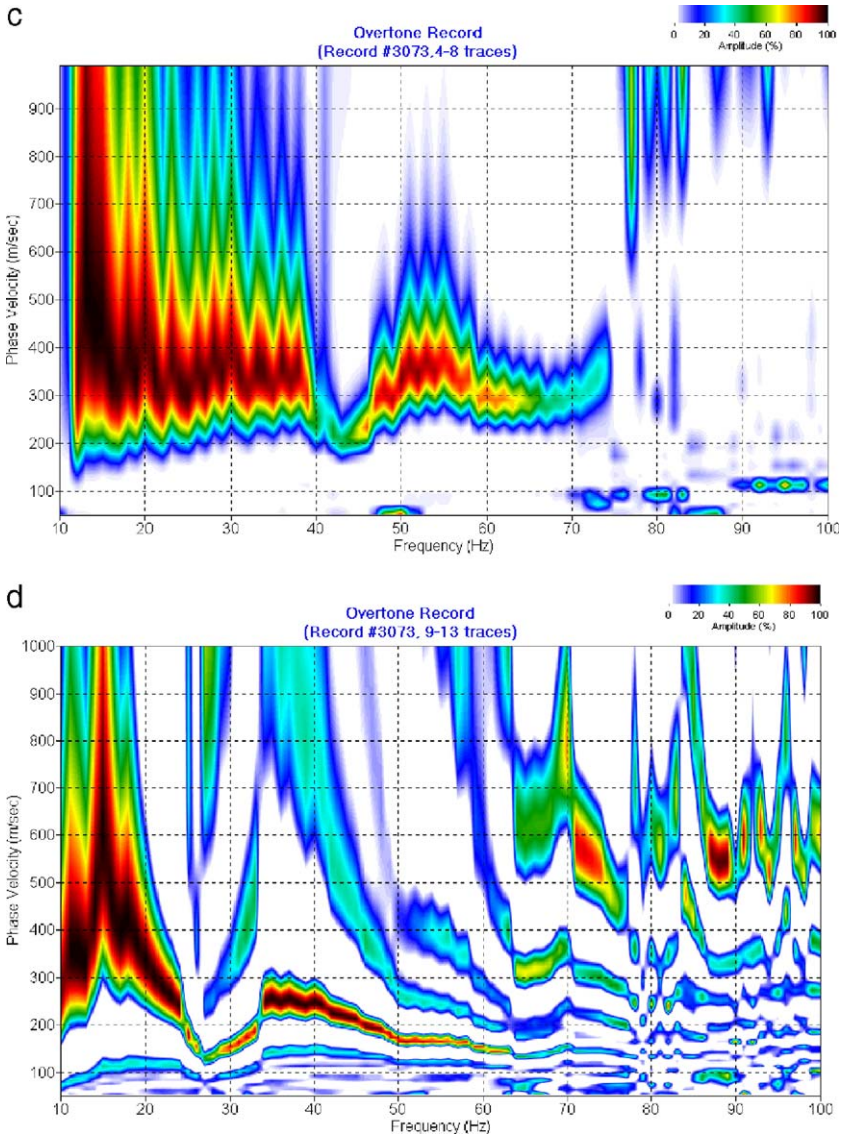

Fig. 2 (*continued*).

modes appeared and can be separated from the fundamental mode from the dispersive image of Fig. 2b, in which body-wave energy did not appear. The geophone spreads for both data sets used to generate images in Fig. 2a and b are the same. The only difference between these data is the change of the nearest offset from 0.6 to 6 m. These observations suggest that the near-field traces, which lack higher mode Rayleigh wave energy and possess a high percentage of body-wave energy, strongly affect the results of dispersion analysis.

To examine in detail how different properties of surface waves change with offset in the near field, traces 4–8 and 9–13 were extracted for analysis (Fig. 2c and d). It is obvious that properties of Rayleigh waves change rapidly with offset in the near field. The problem associated with the nearest offset is not only associated with shot # 3073. We can identify this problem from other shots along the survey line. For example, the same phenomenon is prevalent on shot # 3077 studied here using the same processing parameters in dispersion analysis (Fig. 3a and b). There must be some quantifiable characteristics controlling this phenomenon. The question that naturally arises is: how do you choose the optimum nearest offset for near-surface applications of Rayleigh wave imaging? There are two classical models that we will consider to address this problem.

3. Rayleigh waves due to a surface point load on a homogeneous half-space

Resolution of dispersion image in the frequency–velocity domain is proportional to the geophone-spread length (Forbriger, 2003a). Limited data-acquisition resources are normally associated with near-surface applications such as a 24–(or 48–) channel seismograph and a low energy source (a hammer or a weight dropper). Therefore, the nearest offset should be selected as small as possible with considerations of limiting the required area for data acquisition, compromising resolution of dispersion curves with a limited number of recording channels, and minimizing source energy. When the nearest offset is small, the model of the nearest receiver to source laid out on earth surface can be approximated by a homogenous half-space in many cases. Theoretical discussion of Rayleigh waves due to a surface point load on a homogeneous half-space can be found in numerous publications (e.g., Pekeris, 1955; Pekeris and Lifson, 1957; p. 259–330, Aki and Richards, 1980; Ben-Menahem and Singh, 1981). Although the problem was resolved by Pekeris (1955) and improved by Pekeris and Lifson (1957),

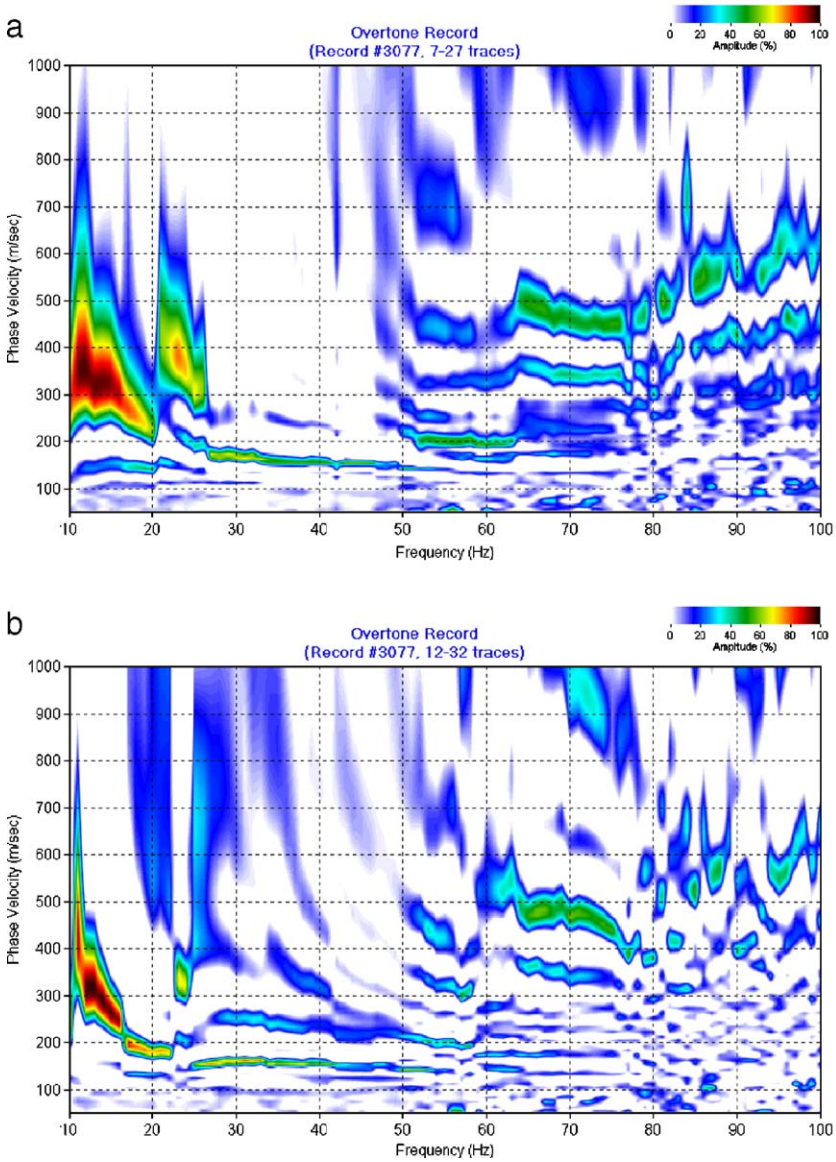

Fig. 3. Dispersion analysis of record #3077 acquired in Joplin, Missouri, in 1997: (a) based on traces from 7 to 27, and (b) based on traces from 12 to 32. Parameters were kept the same during generation of (a) and (b).

here we prefer to use Hanyga's (1985) presentation to begin our discussion because it is straightforward.

The cylindrical coordinate system with vertical z-axe positive downward was adopted. The earth surface was at $z=0$. A point load was located at point (0, 0, 0). Through a complex mathematical derivation, solutions of surface displacements can be obtained by setting two Lame constants equal ($\lambda=\mu$), as:

$$w(\tau)=\begin{cases} 0 & \tau<1/\sqrt{3} \\ \dfrac{M}{32\pi\mu r}\left\{6-\dfrac{\sqrt{3}}{\sqrt{\tau^2-1/4}}-\dfrac{\sqrt{3\sqrt{3}+5}}{\sqrt{\gamma^2-\tau^2}}+\dfrac{\sqrt{3\sqrt{3}-5}}{\sqrt{\tau^2-(3-\sqrt{3})/4}}\right\} & 1/\sqrt{3}<\tau<1 \\ \dfrac{M}{16\pi\mu r}\left\{6-\dfrac{\sqrt{3\sqrt{3}+5}}{\gamma^2-\tau^2}\right\} & 1<\tau<\gamma \\ \dfrac{3M}{8\pi\mu r} & \tau>\gamma \end{cases} \tag{2}$$

$$u(\tau)=\begin{cases} 0 & \tau<1/\sqrt{3} \\ \dfrac{M}{2\pi^2\mu r}\tau R_1(\tau) & 1/\sqrt{3}<\tau<1 \\ \dfrac{M}{2\pi^2\mu r}\tau R_2(\tau) & 1<\tau<\gamma \\ \dfrac{M}{2\pi^2\mu r}\tau R_2(\tau)-\dfrac{M}{8\pi\mu r}\left(\tau^2-\gamma^2\right)^{-1/2} & \tau>\gamma \end{cases} \tag{3}$$

where w is radiation displacement on the horizontal surface, u is vertical displacement; $\tau=V_S t/r$ with t and r representing time and radiation distance, respectively. V_S is velocity of shear wave; M is a constant related to strength of source; $\gamma=V_S/V_R r$ with V_R representing velocity of Rayleigh wave; and R_1 and R_2 are two terms of integration related toτ.

From the equations, one may immediately conclude the following physical interpretations:

a) P-wave is the fastest energy and arrives at $\tau=1/\sqrt{3}$ appearing as jump discontinuities;
b) S-wave is the second energy packet and arrives at $\tau=1$, also appearing as jump discontinuities; and
c) Rayleigh waves are the last to arrive at $\tau=\gamma$ and appear as an infinite pulse dropping immediately to a stationary state that corresponds to a constant force.

Rayleigh waves can be observed after time τ. However τ is apparent time related to true time t by the relation $\tau=V_S t/r$, where $t=r/V_R$. Nothing can be quantitatively determined because of time t and distance r appearing in the relation simultaneously. If the shortest time that Rayleigh waves can be recorded is assumed, such as a period or half of a period of the Rayleigh waves, the nearest offset could be determined from the above relation. However, these kinds of choices are empirical or semi-quantitative and lack the desired rigorous numerical methods.

4. Rayleigh waves to a surface point load on a layered homogenous elastic earth

Having derived nothing numerically useful related to offset for applications in a homogenous elastic half-space, we focus our attention on another commonly used model—a layered homogenous elastic earth. It is well established that the fundamental mode of a Rayleigh wave is not dispersed, with no higher modes in a homogeneous elastic half-space (Lamb, 1904). However the fundamental and higher modes appear simultaneously and become dispersive for a layered elastic model. Dispersion is the result of constructive interference between P and SV waves at the free surface. The dispersion equation (e.g., Lay and Wallace, 1995; Kennett, 2001) gives the velocity of Rayleigh waves as a function of frequency. There are an infinite number of velocity–frequency solutions. Fortunately modes can be separated into two types or branches, symmetric (M1) and antisymmetric (M2). For symmetric modes of a two-layer model (a layer on the top of the half space), vertical displacement has opposite signs at the free surface and interface within the half-space. Antisymmetric modes have the same sign (cf., Udias, 1999).

As stated by Hanyga (1985), M1 branch is characterized by retrograde elliptic particle motion on the surface, and M2 branch is characterized by prograde elliptic particle motion. Based on analysis of the two-layer model by Ben-Menahem and Singh (1981), energy of M21 (the first mode of M2 branch) is concentrated mainly in the vicinity of the interface, whereas the energy of M11 (the first mode of M1 branch, i.e., the fundamental mode) is concentrated predominately near the free surface. In the limit of low frequencies (the cut-off frequency corresponding to wavenumber $k\rightarrow 0$), the velocity of M11 tends toward the Rayleigh wave velocity that corresponds to the half space, whereas for all other modes, velocities at the cut-off frequencies tend to the S-wave velocity of the half space. In the limit of high frequencies ($k\rightarrow\infty$), the velocity of M11 tends to the Rayleigh wave of the upper layer, whereas velocities of the other modes approach the S-wave velocity of the upper layer.

Based on the conventional geometrical ray theory, for a two-layer model interference of P and S waves can take place everywhere on the free surface except at the source point in all but the following two cases. The first is when the SV-wave propagates along the free space at a lower velocity and therefore can meet P-wave reflections from the interface with a higher velocity. The second is when reflected SV energy from the interface occurs on the free surface at an angle in excess of the

critical angle ($i_{SV} = \sin^{-1}\frac{V_s}{V_p}$). This scenario yields evanescent P-waves and reflected SV-waves propagating together along the free surface (cf., Lay and Wallace, 1995, p. 120).

From an application perspective, attention must be focused on where dispersed Rayleigh waves can be well developed and recorded, rather than on how they form. With that in mind, let us consider the shallowest interface under the ground surface as our target. The source is located at point A (Figs. 4 and 5). The velocities of P and S waves in the upper layer are marked as V_p and V_s.

4.1. Case I

In one case (Fig. 4), SV-waves propagating along the free surface meet reflected P-wave energy causing interference. Dispersed Rayleigh waves form as a result of their interference at the free surface. The shortest distance d that this energy can be observed can be determined using the following relations:

$$\frac{\overline{CB} + \overline{AC}}{V_p} = \frac{d}{V_s}, \ \overline{CB} = \overline{AC} = h/\cos i, \text{ and} \tag{4}$$

$$\tan i = d/2h. \tag{5}$$

Joining Eqs. (4) and (5), we have $\sin i = V_s/V_p$, and so

$$\cos i = \sqrt{1 - \sin^2 i} = \sqrt{1 - V_s^2/V_p^2}. \tag{6}$$

Inserting Eq. (6) into Eq. (4), we obtain

$$d = \frac{2h}{\sqrt{\sigma^2 - 1}}, \tag{7}$$

where $\sigma = V_p/V_s$.

4.2. Case II

In the other scenario, point B is assumed the closest position where dispersed Rayleigh waves can be

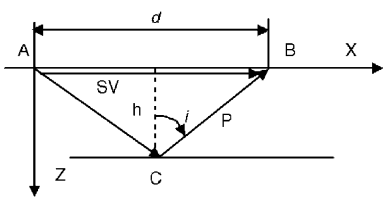

Fig. 4. Dispersed Rayleigh waves occur where interference between SV-waves that propagate along the free surface and the P-wave energy reflected from an interface.

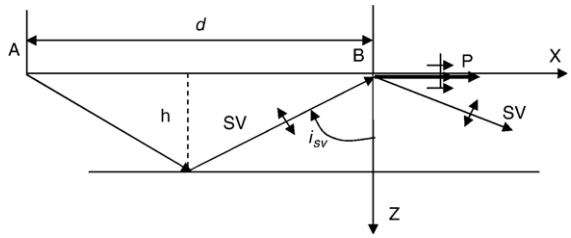

Fig. 5. The second explanation of the nearest distance where dispersed Rayleigh waves occur for a planar layered elastic earth model (small arrows represent particle oscillations. Other symbols have the same meanings as Fig. 4).

recorded. From a simple geometric relation (Fig. 5), the closest distance from the source can be determined by the following relationship:

$$d = 2h \tan i_{sv}, \tag{8}$$

where h is the depth of the point where SV energy is reflected at the critical angle. Inserting the critical angle $i_{SV} = \sin^{-1}\frac{V_s}{V_p}$ into Eq. (8), one immediately has

$$d = 2h\frac{V_s}{\sqrt{V_p^2 - V_s^2}} = \frac{2h}{\sqrt{\sigma^2 - 1}}, \tag{9}$$

which is the same solution as Case I.

Eqs. (7) or (9) provides the lower limit of the nearest offset in acquiring Rayleigh-wave data. For example, assuming Vp/Vs is 4, which is common for near-surface materials, the nearest offset d is approximately equal to $h/2$ that indicates Rayleigh waves would appear where the offsets are larger than one-half of

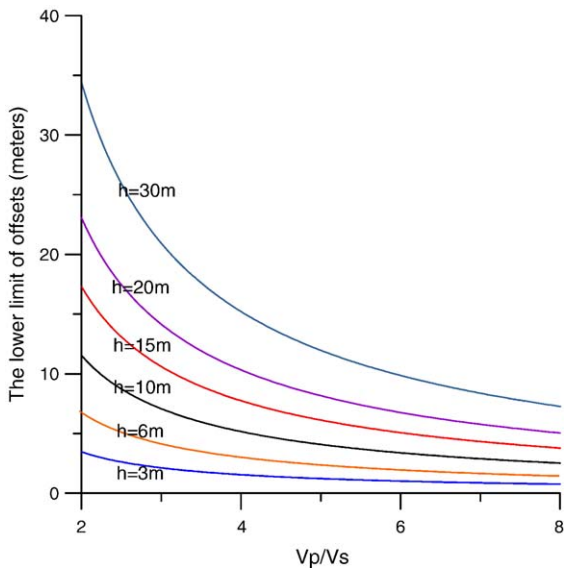

Fig. 6. The nearest offset varies with the ratio of V_p/V_s and the depth to the shallowest interface.

the thickness of the first layer (Fig. 6). No geophone should be placed where the offset is less than the offset determined by Eq. (7) using a priori geological information and velocities. It saves valuable resources (not losing near-offset traces) in near-surface investigations.

As an example, with $h=10$ m, $V_p=600$ m/s, $V_s=200$ m/s, and using Eqs. (7) or (9), the theoretical offset is 7.1 m. The result is reasonable from previous empirical studies and within the range of Table 1.

Studying the experimental records previously discussed, seismic velocities are 1100 m/s for P-wave determined from the direct wave of records and 220 m/s for S-wave determined by dispersion analysis, respectively. Theoretically, the offset should be 4.9 m based on Eq. (7). This calculated offset agrees well with the improved dispersion image with the nearest offset of 6m (Fig. 2b). Near-field effects were obviously suppressed. This observation means traces 4–8 were beyond the lower limit of offset and therefore show strong near-field effects on resulting dispersion images.

5. Discussion and conclusions

In practice, a layered homogeneous elastic model is a useful approximation for many near-surface situations in which some P- and S-wave velocities in the uppermost layer can be measured accurately. Eqs. (7) or (9) discussed in the paper can be used to guide selection of the nearest offset for surface wave surveys on shallow subsurface investigations. However, it should be noted that the lower limit of the nearest offset calculated from the formulas given in this paper is not based on resolution or energy spectra of the records. Approaches described here should be used for studies where dispersive Rayleigh waves are generated by a point source on the earth surface above a layered homogeneous elastic earth.

This paper focused the discussion on the nearest offset for Rayleigh-wave surveys that investigate near-surface materials. Although the approach is discussed based on a two-layer model, it is valid for a multi-layer model because the reflected P-wave energy from the first interface always interferes with the direct S-wave energy first. Therefore, the nearest offset determined by Eq. (7) is applicable for a multi-layer model. It is true for real-world applications that the selection of the nearest offset may depend on the depth of investigation (wavelength) without information on near-surface velocities as several researches suggested (e.g., Rix and Leipski, 1991; Stokoe et al., 1994; Xia et al., 2004a). However, it possesses a risk to lose high-frequency components of Rayleigh waves that are critical to define top layers of the subsurface. If the main interest is deeper layers and resolution of the top layers is not crucial, an averaged velocity ratio of top layers may be used in Eq. (7) to define the nearest offset.

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